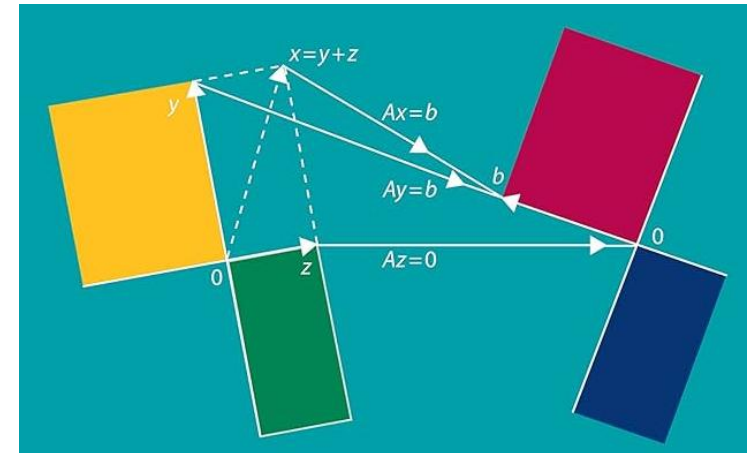


Introduction to Linear Algebra

(Linear Algebra)

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Who am I...

- *BSc in Engineering Physics* from the University of Colombo, Sri Lanka in 2006.
- *Bachelor of Information Technology (BIT) (External)*, UCSC in 2009.
- *Research Assistant*, Language Technology Research Laboratory (LTRRL) (2008 – 2018)
- *PhD in Computer Science* from the University of Colombo, Sri Lanka, 2017.
- *Principal Investigator*, Language Technology Research Laboratory (LTRRL), UCSC.

Research Interests:

- *Natural Language Processing/Human Language Technology - Machine Translation, Speech Processing (ASR and TTS) and Machine Learning, Conversational Agents (Chatbots), Large Language Models (LLMs).*

- Addresses the need for local language computing in Sri Lanka through research and development in Localization and Language Processing
- Corpus Creation (Dataset Creation) - *Sinhala and Tamil*
- Voice (Speech) to Text System (ASR) – *Sinhala*
- Text to Voice (Speech) System (TTS) – *Sinhala and Tamil*
- Machine Translation System – *Sinhala – Tamil Language Pair*
- AI Chatbot Systems – *English and Sinhala (with Bank of Ceylon)*
- Spell Checking Application – *Sinhala*
- Image to Text Conversion System (OCR) – *Sinhala and Tamil*
- Grammar Checker– *Sinhala*

Course Organization

Lecturer: Dr. Randil Pushpananda

Assistant Staff: Mr. Dilum Gamage
Ms. Shashini Wijewardhana

Lectures – Friday 08.00 AM – 10.00 AM

Practicals – Wednesday 01.00 PM – 03.00 PM

Course Organization

Recommended Reading:

- Strang, G., 2022. Introduction to linear algebra
- Friedberg, S.H., Insel, A.J. and Spence, L.E., 2018. *Linear algebra*
- Jim Hefferon, Linear Algebra

Evaluation Criteria :

80% paper, 20% Assignments

Assignments:

- 2 In-Class Assignments

Course Outline

Number of Credits	2L + 1P (3 Credits)	
Core / Optional	Core	
Evaluation Criteria	Assignments:	20%
	Final Exam:	80%

Method of Delivery		Per Week	Total
	Lectures	2 Hours	30 Hours
	Tutorials	2 Hours	30 Hours
	Lab Work	-	-
	Group Work	-	-

Course Contents

1. Linear Systems
2. Vector spaces and subspaces
3. Orthogonality
4. Determinants
5. Eigenvalues and Eigenvectors
6. Singular Value Decomposition (SVD)
7. Linear Transformations
8. Applications of Linear Algebra in Various Disciplines

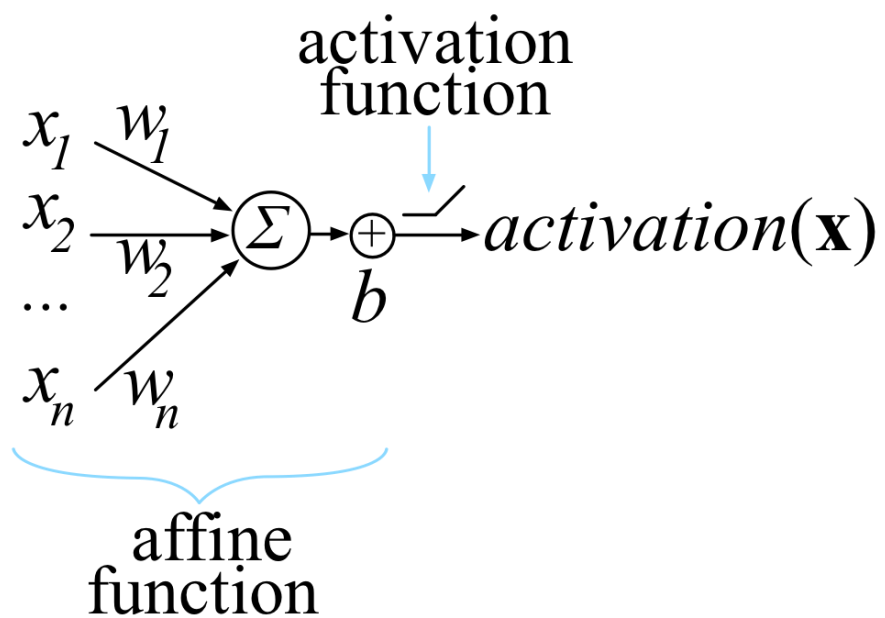
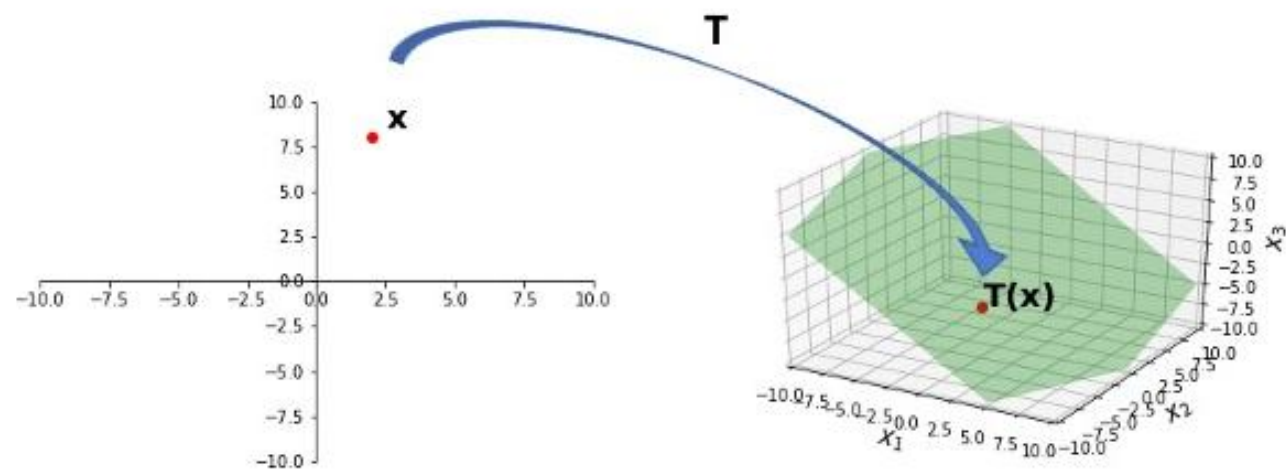
What is Linear Algebra?

- Linear algebra is a branch of mathematics that deals with vector spaces and linear mappings between these spaces.
- Includes various concepts such as vectors, matrices, systems of linear equations, determinants, eigenvalues, and eigenvectors.
- Linear algebra finds extensive applications in diverse fields including physics, engineering, computer science, economics, and statistics.

Why?

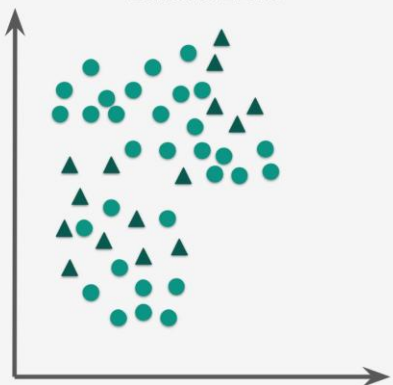
Why?

- Foundational Understanding:
 - Key mathematical concepts such as vectors, matrices, and linear transformations.
 - Many algorithms and techniques used in computer science are based on mathematics.

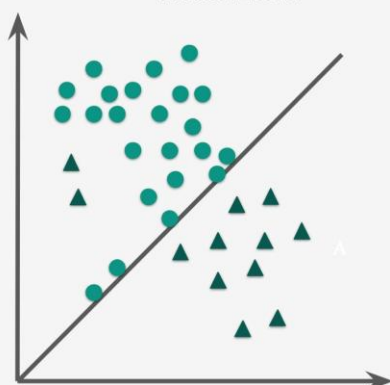


Linear Discriminant Analysis

Before LDA



After LDA



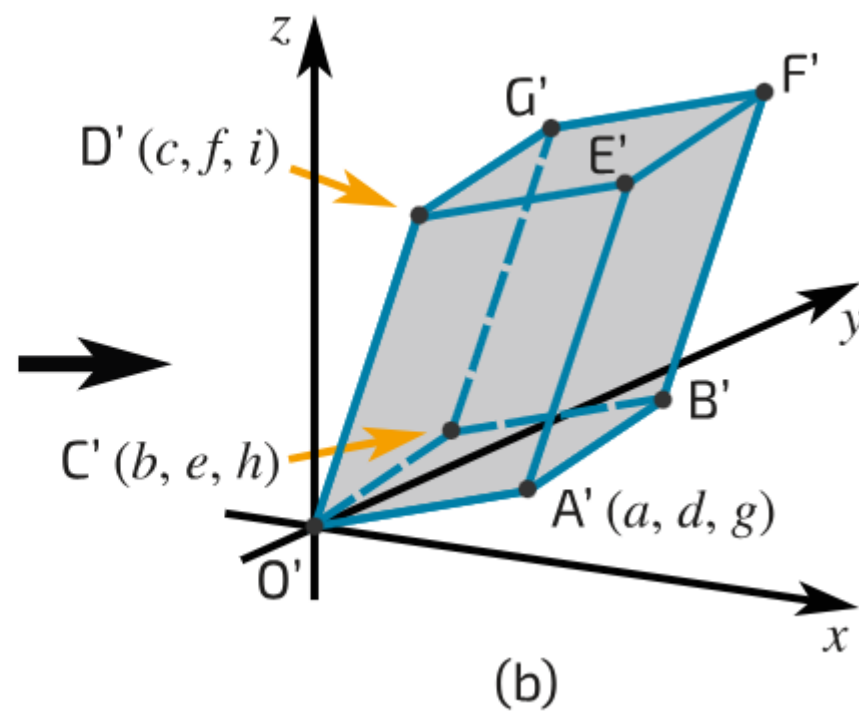
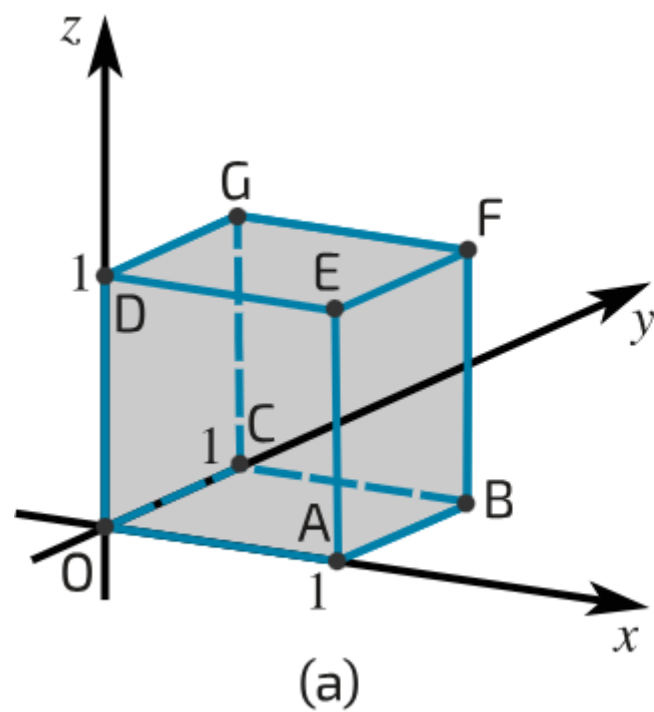
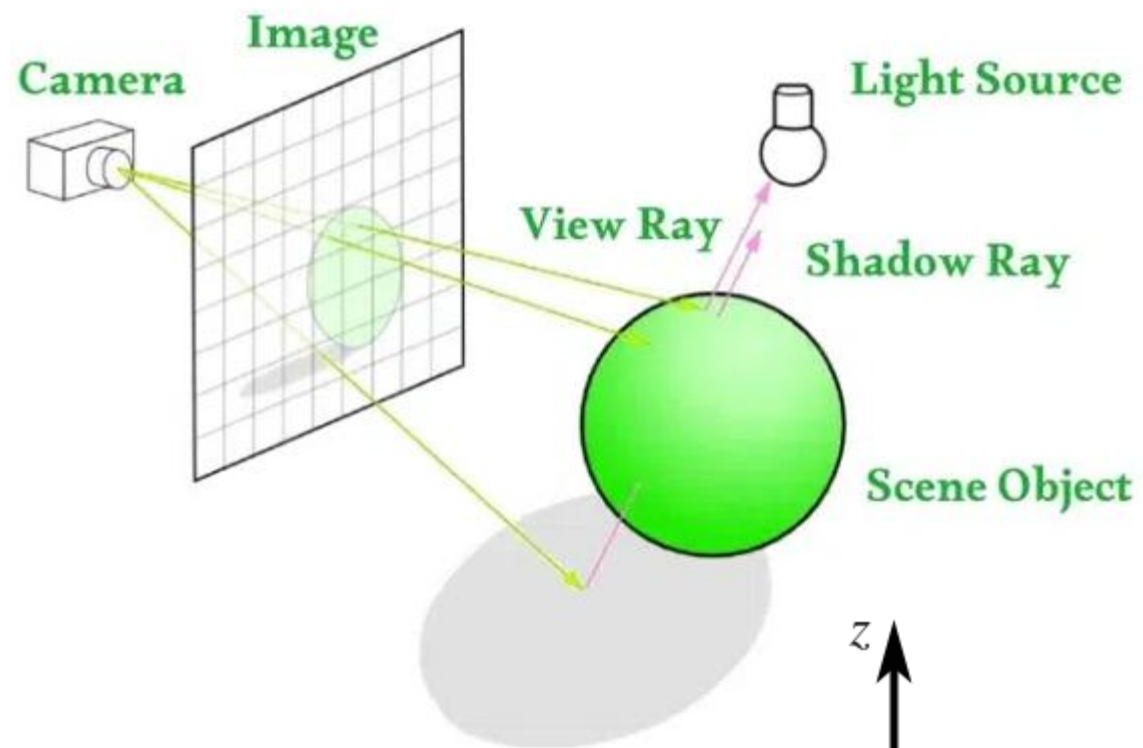
Linear Transformation

$$\begin{bmatrix} 1/2 & -1/2 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1\frac{1}{2} \\ 0 \end{bmatrix}$$

Why?

- Graphics and Computer Vision:

- Linear algebra is crucial in graphics programming for tasks such as rendering 3D scenes, transformations, and lighting calculations.
- In computer vision, techniques like image processing, feature extraction, and object recognition heavily rely on linear algebra.



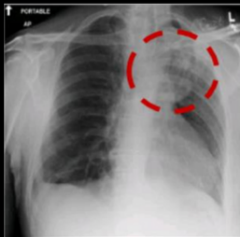
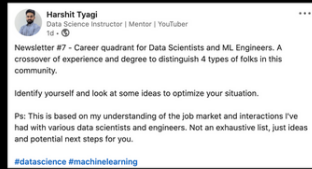




Why?

- **Machine Learning and Data Science:**
 - Backbone of many machine learning algorithms, including regression, dimensionality reduction techniques like neural networks.
 - Understanding concepts like matrices, vectors, and matrix operations is essential for implementing and understanding these algorithms.
- **Data Representation and Manipulation:**
 - Data in computer science is often represented and manipulated using matrices and vectors.
 - Linear algebra provides tools for organizing, transforming, and analyzing data efficiently, making it indispensable in fields like databases, computer graphics, and cryptography.

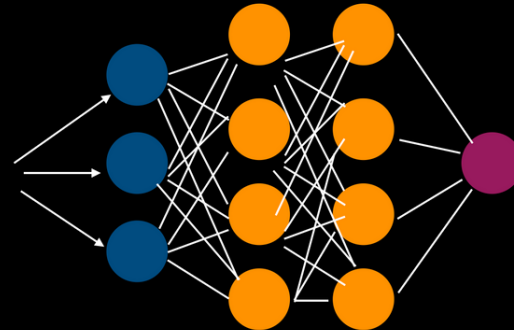
ML models Leveraging Linear Algebra



[[12, 213, 44, 15],
[20, 113, 39, 10],
[13, 117, 45, 13],
[25, 120, 49, 14],

...,
...,
]

Tensor



[[0.98, 0.45, 0.44, 0.95],
[0.23, 0.13, 0.39, 0.89],
[0.13, 0.77, 0.55, 0.23],
[0.25, 0.20, 0.49, 0.14],

...,
...,
]

Tensor

Attend the session!

Atelactasis

Ok Google, set alarm
for 5 AM.

Linear algebra = language of modern AI

Introduction to Linear Algebra

- Questions:

- Consider equation $4x = 8$,

Find x ?

Introduction to Linear Algebra

- Questions:

- Saman has bought two ice creams and two drinks for Rs. 300.00. How much did John pay for each item?

Introduction to Linear Algebra

- Questions:

- Saman has bought two ice creams and two drinks for Rs. 300.00. Kamala bought two ice creams and one drink for Rs. 200.00. How much did they pay for each item?

To extract information from multiple linear equations, we need linear algebra

Complex scientific, or engineering problem can be solved by using linear algebra on linear equations

Linear Equation

- A linear equation is one where all the variables such as x , y , z have index (power) of 1 or 0 only

$$x + 2y + z = 5$$

$$x = 3$$

$$3x + y + z + w = -8$$

Linear Equation

$$x + 2y + 3z = 5$$

Variables: x, y, z

Coefficients: 1, 2, 3

Constant: 5



Fundamental Problem of Linear Algebra

Solving Systems of Linear Equations

Questions

Which of the following equations are Linear Equations

- (a) $x - y - z = 3$ (c) $\cos(x) + \sin(y) = 1$ (b) $\sqrt{x} + y + z = 6$
(d) $e^{x+y+z} = 1$ (f) $x = -3y$ (e) $x - 2y + 5z = \sqrt{3}$

Geometric Representation

- Each equation = a line
- Solution = intersection point
- 3 cases:
 - One solution (intersect)
 - No solution (parallel)
 - Infinite solutions (same line)

Vectors and Matrices

Vectors

- Vectors are one of the most important ideas in linear algebra because they are used to represent:
 - quantities
 - directions
 - data
 - relationships between variables.
- Linear algebra is fundamentally the study of vectors and transformations of vectors.

Vectors Help Describe Geometry

- Vectors help us understand:
 - lines, planes, distances, directions, projections, transformations.
- Example:
 - computer graphics,
 - robotics,
 - gaming.

Vectors Are the Foundation of Machine Learning

- Modern AI depends heavily on vectors.

- Examples:

- Neural network inputs
- Word embeddings
- Feature vectors
- Recommendation systems

- Without vectors:

- no deep learning,
- no search engines,
- no modern data science.

Vectors Allow Efficient Computation

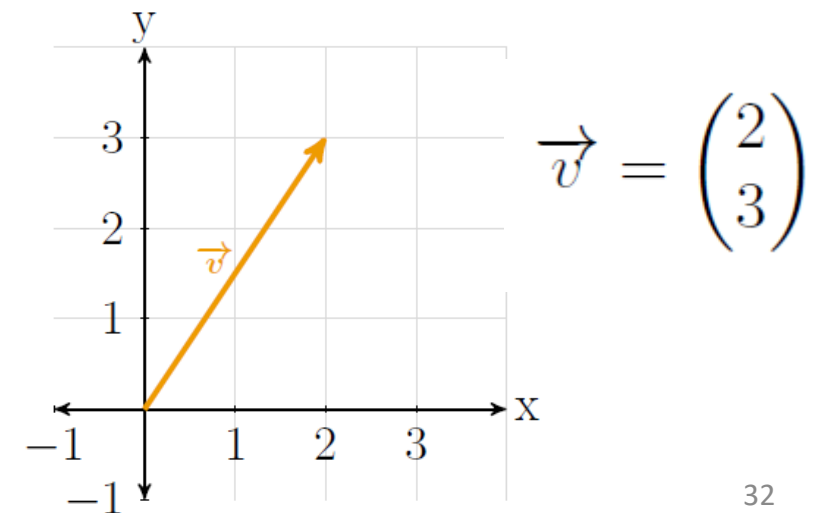
- Computers process large data efficiently using vector and matrix operations.
- Applications:
 - image processing,
 - databases,
 - cryptography,
 - optimization.

Vectors

- Vector is a quantity that has size (magnitude) and direction
 - walk due north for 5 kilometers
 - Velocity, acceleration, force and displacement are all vector quantities
- Scalar is a number that measures the size of a particular quantity.
 - Length, area, volume, mass and temperature are all scalar quantities

In a 2-dimensional space, a vector can be written as $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$.

In a 3-dimensional space, a vector can be written as $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$.



Vector Addition

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{4} \end{bmatrix}$$

$$\mathbf{v} + \mathbf{w} =$$

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \mathbf{1} \\ \mathbf{3} \end{bmatrix}$$

Scalar Multiplication

- Scalar can be a real number. $\lambda \in \mathbb{R}$

- If $\lambda > 1$ the vector will keep the same direction but stretch.

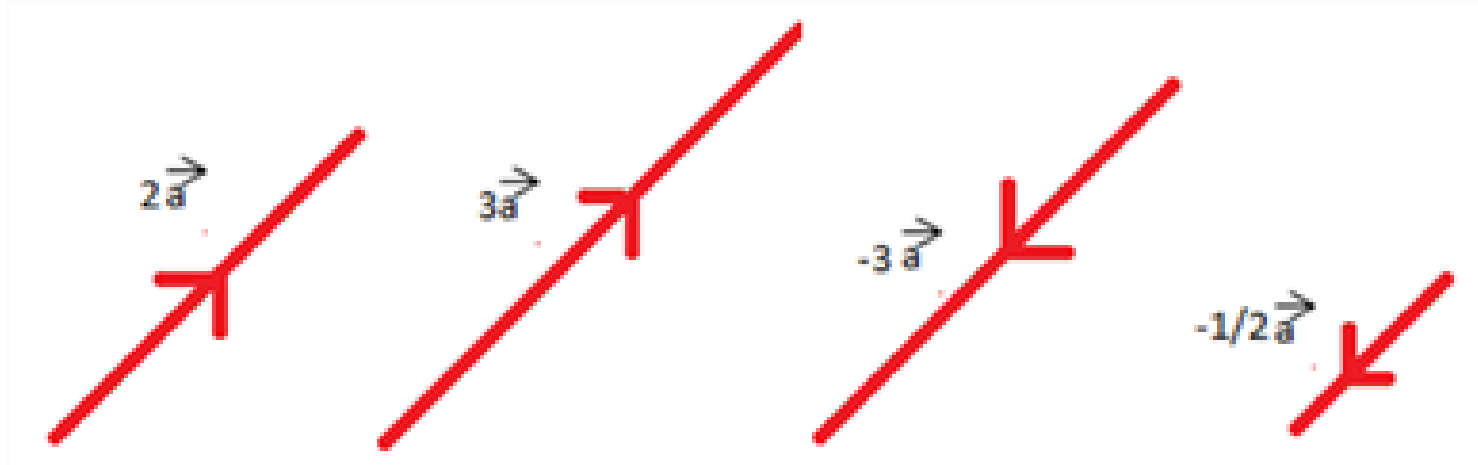
- If $1 > \lambda > 0$ the vector will keep the same direction but shrink.

- If $\lambda < -1$ the vector will change direction and stretch.

- If $0 > \lambda > -1$ the vector will change direction and shrink.

Scalar Multiplication

$$3\mathbf{v} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$



Linear Combination

- The heart of Linear Algebra is in two operations: Addition and Multiplication
- Combining those two operations give Linear Combination

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \mathbf{v} + \mathbf{w} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

The linear combinations of $\mathbf{v}$ and $\mathbf{w}$ are the vectors $c\mathbf{v} + d\mathbf{w}$ for all numbers c and d :

$$\text{The linear combinations} \quad c \begin{bmatrix} 2 \\ 4 \end{bmatrix} + d \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2c + 1d \\ 4c + 3d \end{bmatrix} \quad \text{fill the } xy \text{ plane}$$

Linear Combination

All combinations $c \begin{bmatrix} 4 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ fill the xy plane. They produce every $\begin{bmatrix} x \\ y \end{bmatrix}$

The vectors $c \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$ fill a plane in xyz space.

Problem

- Linear algebra is not limited to 2 vectors in 2-dimensional and 3-dimensional space.
- As long as we stay linear, the problems can get more dimensions and more vectors.
- vectors $\rightarrow$ m components instead of 2 components.
- Can have n vectors $v_1, v_2, \dots, v_n$, instead of 2 vectors.
- Finally, n vectors in m -dimensional space will go into the columns of an m by n matrix A :

$$\begin{array}{l} m \text{ rows} \\ n \text{ columns} \\ m \text{ by } n \text{ matrix} \end{array} \quad A = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$$

Problem

m rows
 n columns
 m by n matrix

$$A = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$$

- 1 Describe all the combinations $Ax = x_1v_1 + x_2v_2 + \cdots + x_nv_n$ of the columns**
- 2 Find the numbers x_1 to x_n that produce a desired output vector $Ax = b$**

Solving Two Equations

Solve $c \begin{bmatrix} 2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$

This means

$$\begin{aligned} 2c + 2d &= 8 \\ c - d &= 2 \end{aligned}$$

Column Way, Row Way, Matrix Way

Column way
Linear combination

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + d \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Row way
Two equations for c and d

$$v_1 c + w_1 d = b_1$$

$$v_2 c + w_2 d = b_2$$

Matrix way
2 by 2 matrix

$$\begin{bmatrix} v_1 & w_1 \\ v_2 & w_2 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Column Picture vs Row Picture

- Column Picture and Row Picture are two different ways to interpret the same system of linear equations.
- Both describe the same equation: $Ax = b$ and produce the same solution set.
- **Row picture** views the system as a collection of equations. (Row picture = looking at the problem from the “equation side”)
- **Column picture** views the system as a combination of column vectors. (Column picture = looking from the “vector construction side”)

Question 1

$$2x - y = 0$$

$$-x + 2y = 3$$

- Find
- (i) Row Picture
 - (ii) Column Picture
 - (iii) Matrix Form

Question 2

$$2x + y = 3$$

$$x - 2y = -1$$

- Find
- (i) Row Picture
 - (ii) Column Picture
 - (iii) Matrix Form

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + d \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

If the points v and w and the zero vector 0 are not on the same line, there is exactly one solution c, d . Then the linear combinations of v and w exactly fill the xy plane. The vectors v and w are “linearly independent”. The 2 by 2 matrix $A = \begin{bmatrix} v & w \end{bmatrix}$ is “invertible”.

Zero Vector

Linearly Independent

Invertible

Zero Vector

- Vector that has a zero magnitude and no direction.
- Example:
 - Suppose two people are pulling a rope from its two ends with equal force but in opposite directions. So, the net force applied to the rope will be a zero vector (null vector) as the two equal forces balance each other out because they are in opposite directions

Linearly Independent

- Two or more vectors are said to be linearly independent if none of them can be written as a linear combination of the others.

$$\begin{bmatrix} 1 \\ 1 \\ 3 \\ 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ -1 \\ -3 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 9 \\ 6 \end{bmatrix} \xrightarrow{\text{linear combination}} c_1 \begin{bmatrix} 1 \\ 1 \\ 3 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \\ -1 \\ -3 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 5 \\ 9 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Invertible

- A square matrix such that the product of the matrix and its inverse generates the identity matrix.

$$AB = BA = I_n$$

$$\Rightarrow B = A^{-1}$$

Where,

A is $(n \times n)$ invertible matrix.

B is $(n \times n)$ matrix called inverse of A .

I_n is $(n \times n)$ identity matrix.

Vectors in 3 - Dimensions

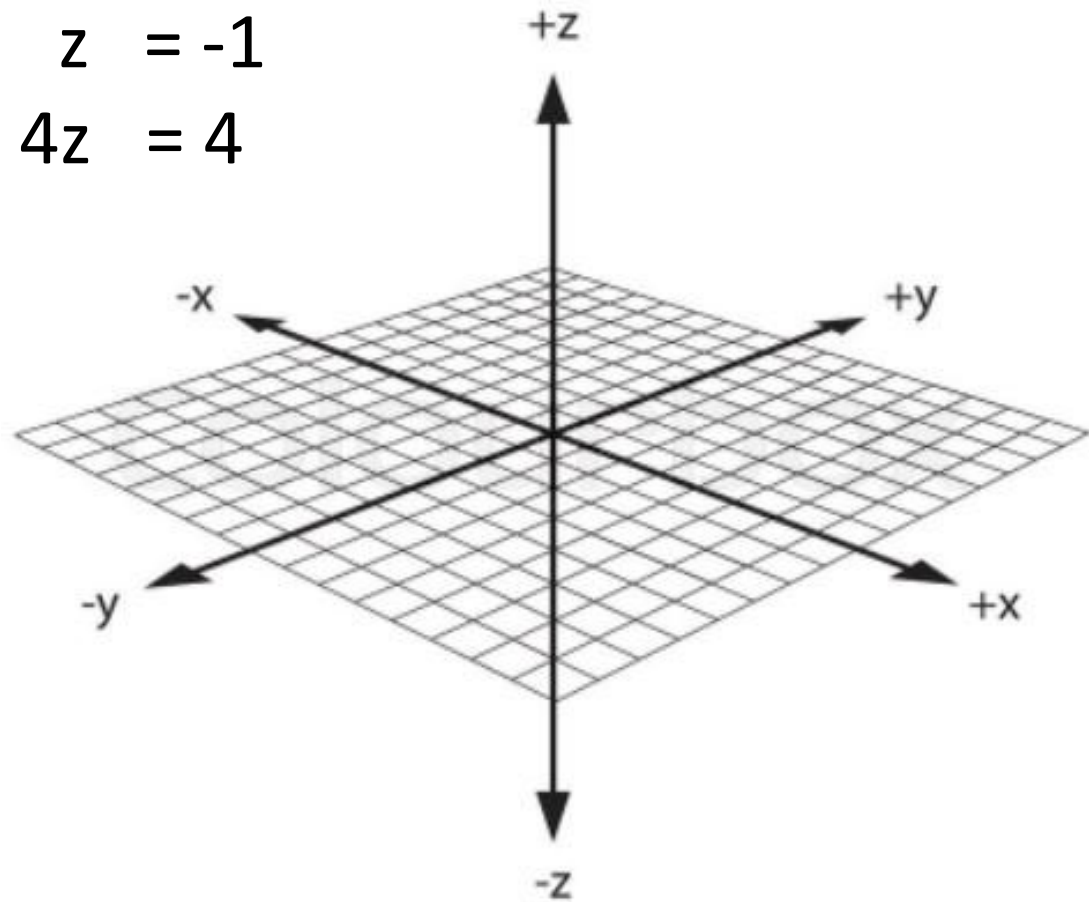
$$\begin{array}{rcl} 2x & - & y & = & 0 \\ -x & + & 2y & - & z & = & -1 \\ & & -3y & + & 4z & = & 4 \end{array}$$

Find x, y and z

Vectors in 3 - Dimensions

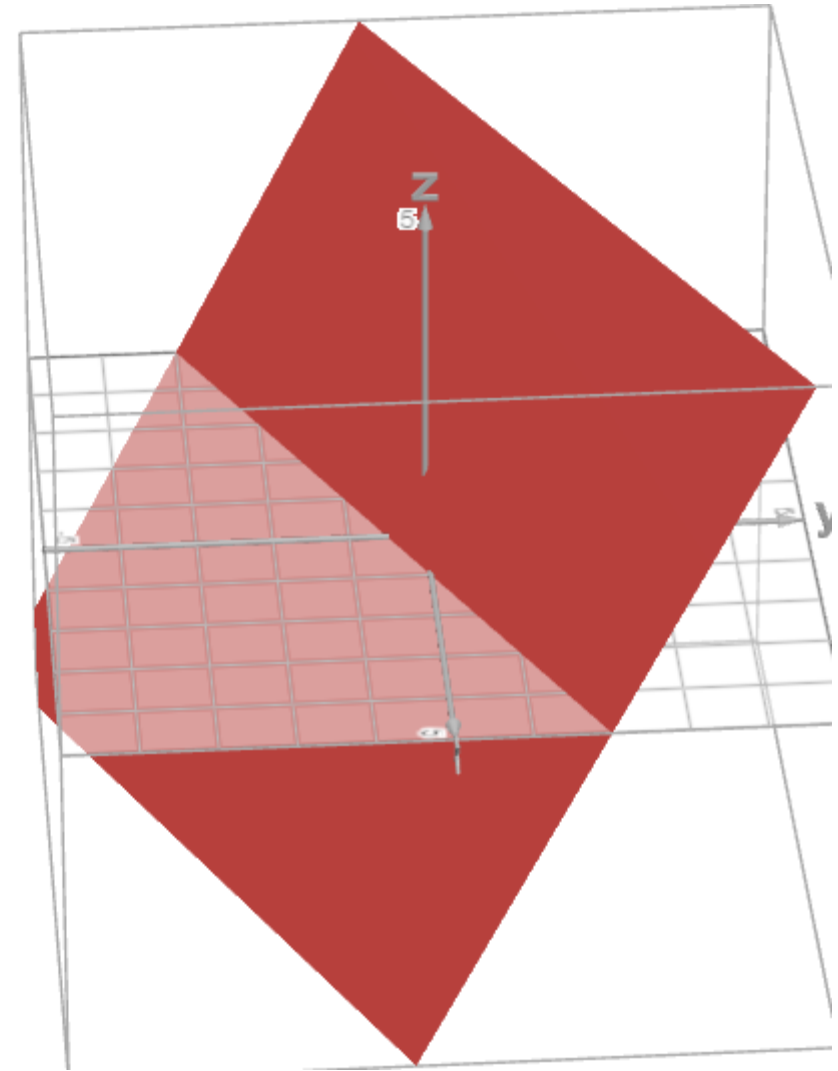
$$\begin{array}{rcl} 2x & - & y & & = & 0 \\ -x & + & 2y & - & z & = & -1 \\ & & -3y & + & 4z & = & 4 \end{array}$$

Row Picture



Try: <https://www.desmos.com/3d>

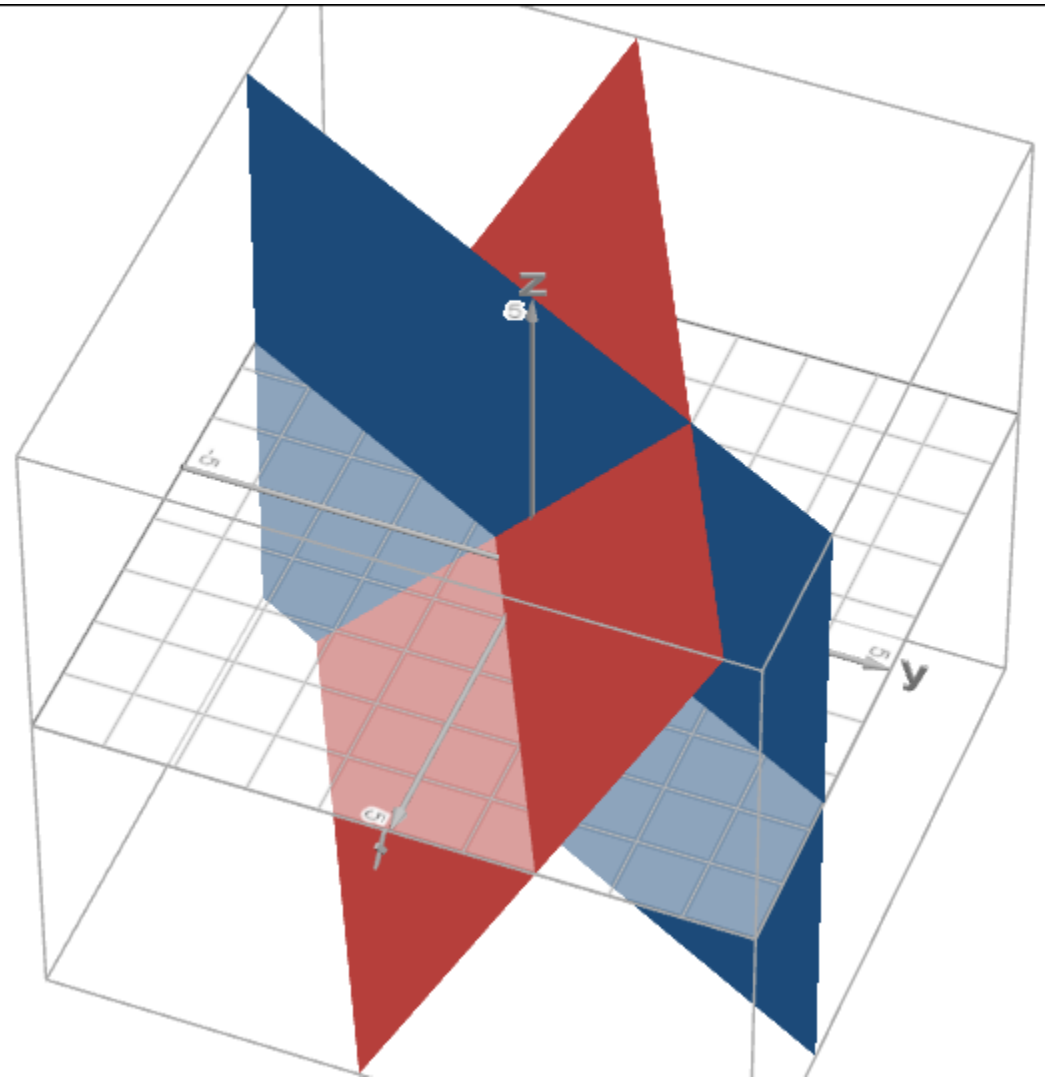
$$-x + 2y - z = -1$$



$$-x + 2y - z = -1$$

$$2x - y = 0$$

☒ Extend to 3D

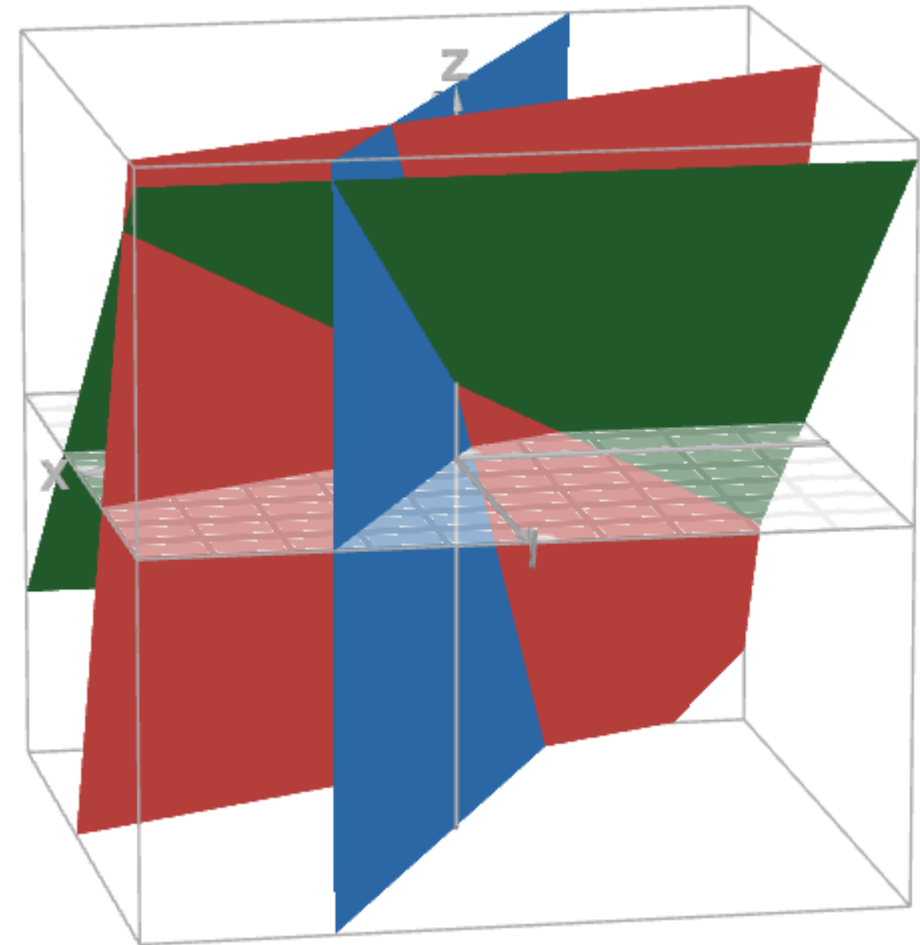


$$-x + 2y - z = -1$$

$$2x - y = 0$$

☒ Extend to 3D

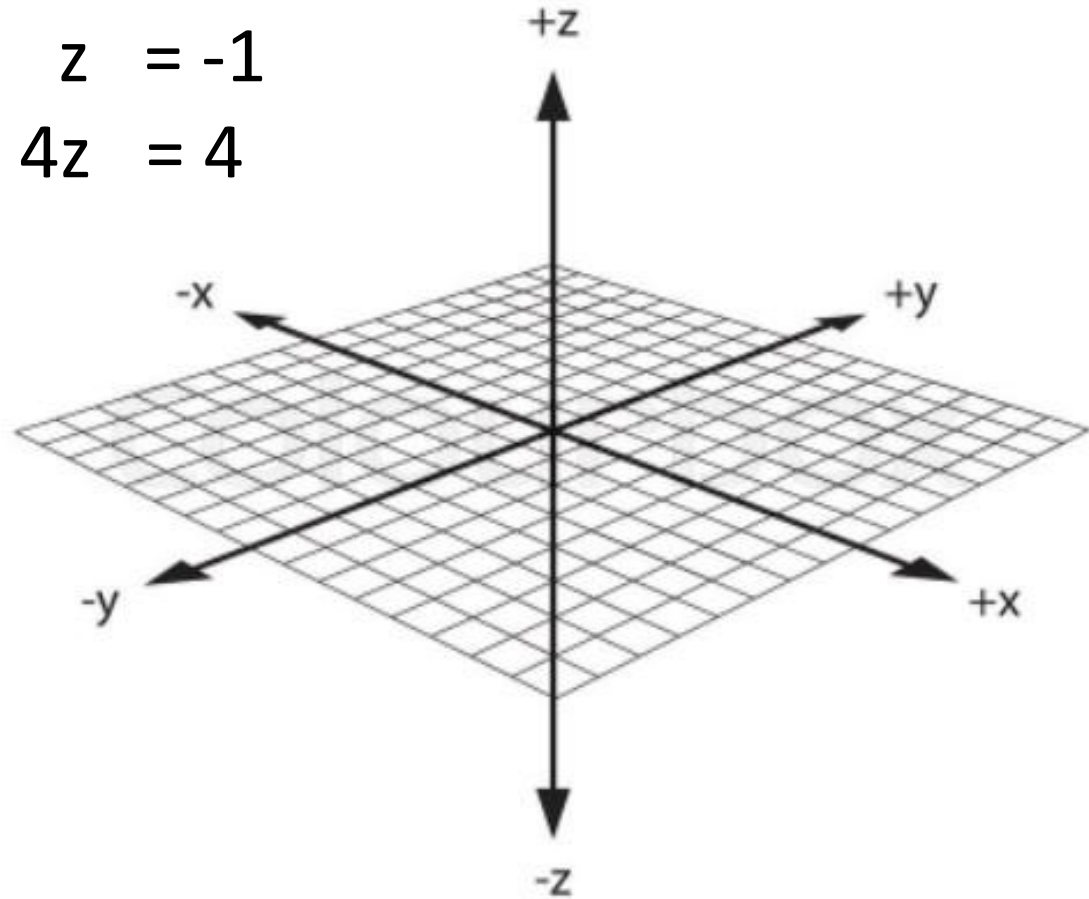
$$-3y + 4z = 4$$



Vectors in 3 - Dimensions

$$\begin{array}{rcl} 2x & - & y & & = & 0 \\ -x & + & 2y & - & z & = & -1 \\ & & -3y & + & 4z & = & 4 \end{array}$$

Column Picture

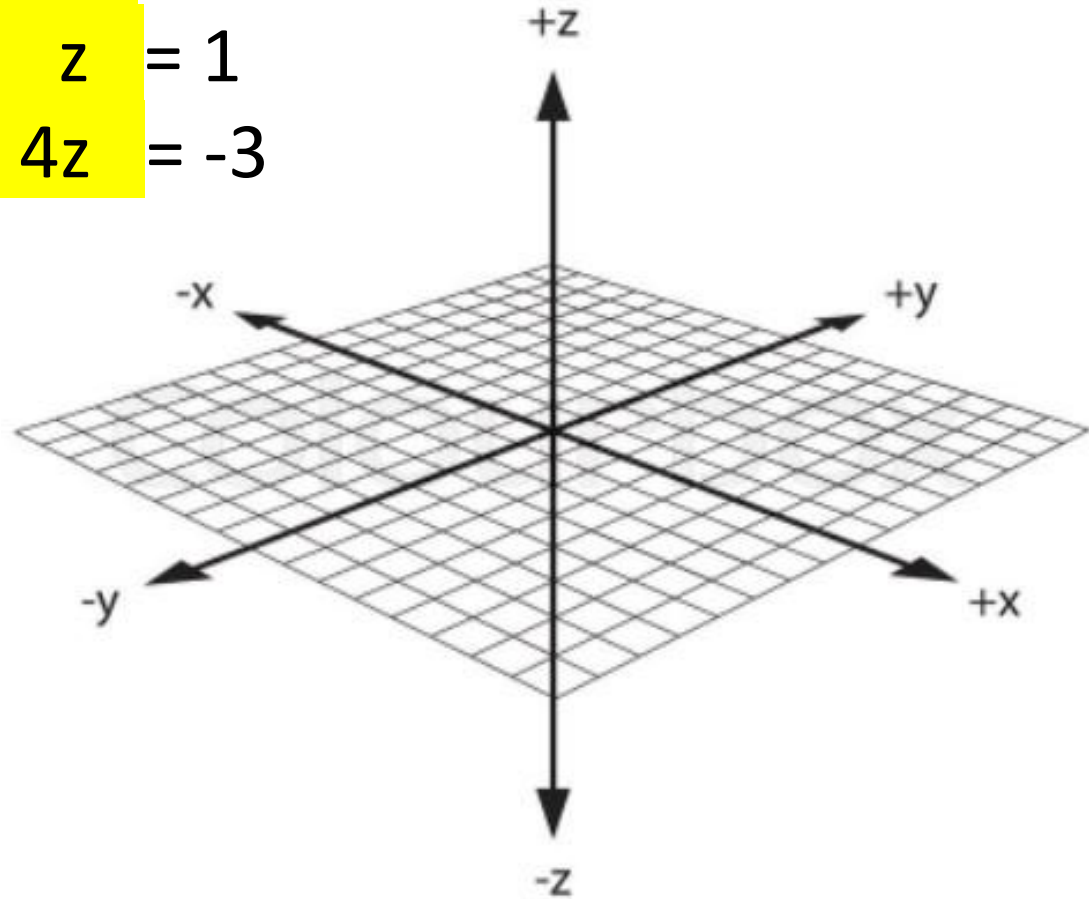


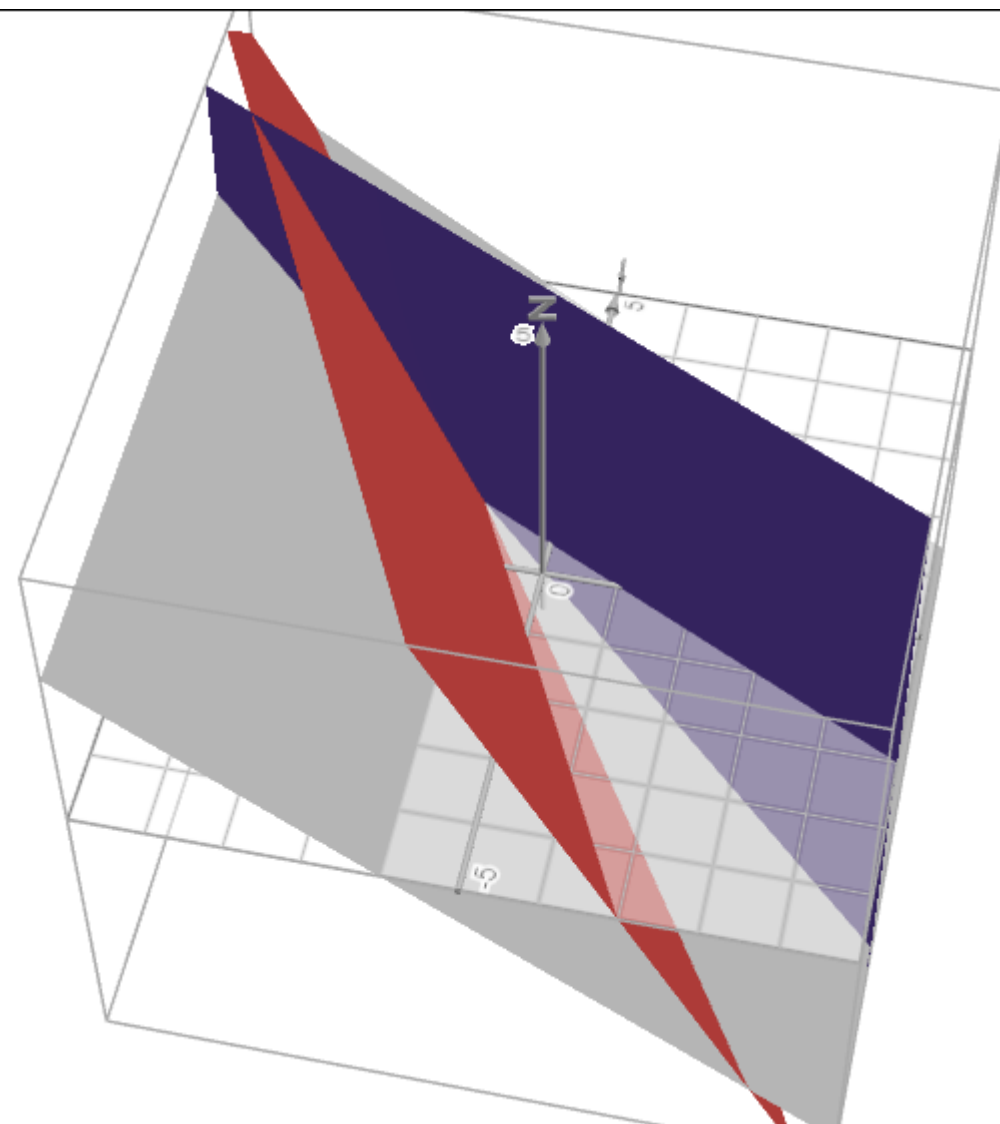
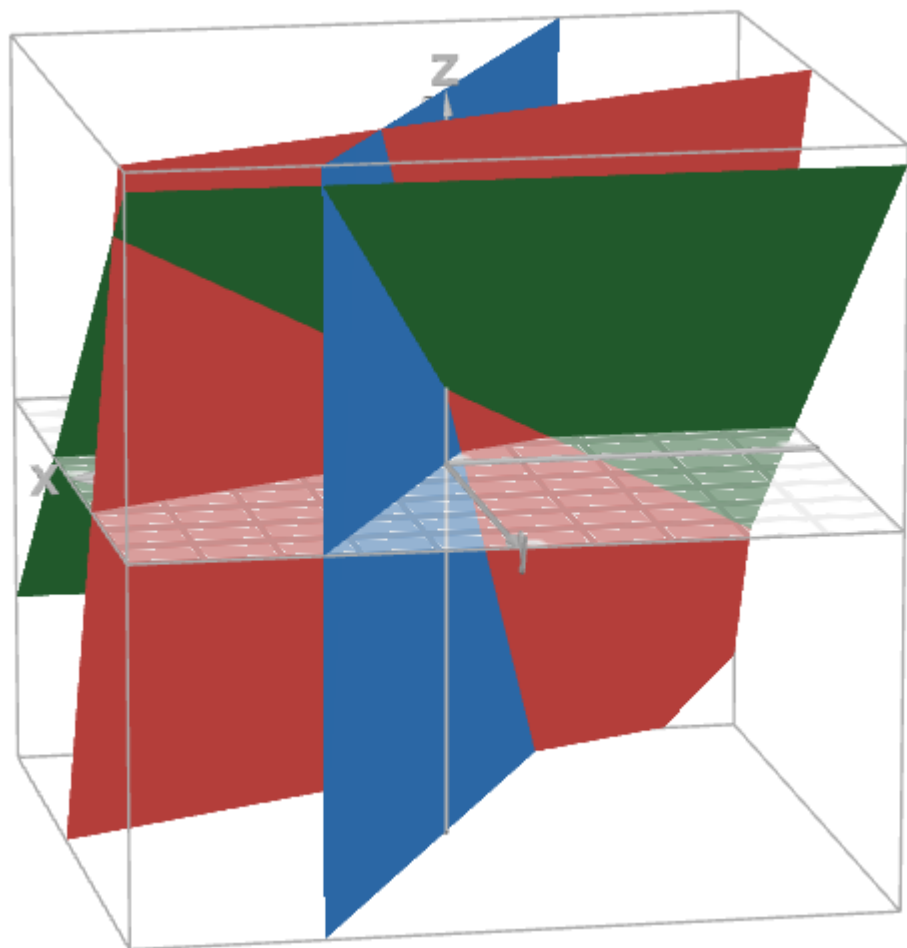
Vectors in 3 - Dimensions

$$\begin{array}{rcl} 2x & - & y & & = & 1 \\ -x & + & 2y & - & z & = & 1 \\ & & -3y & + & 4z & = & -3 \end{array}$$

Column Picture

$x = ?$, $y = ?$, $z = ?$





Matrix Form

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} \qquad \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$$

$A.x = b$



Matrices and Their Column Spaces

1 $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ is a **3 by 2 matrix**: $m = 3$ rows and $n = 2$ columns. Rank 2.

2 The 3 components of $A\mathbf{x}$ are dot products of the 3 rows of A with the vector $\mathbf{x}$:

$$\begin{array}{l} \text{Row at a time} \\ A \text{ times } \mathbf{x} \end{array} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 7 + 2 \cdot 8 \\ 3 \cdot 7 + 4 \cdot 8 \\ 5 \cdot 7 + 6 \cdot 8 \end{bmatrix} = \begin{bmatrix} 23 \\ 53 \\ 83 \end{bmatrix}.$$

3 $A\mathbf{x}$ is also a combination of the columns of A $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} + 8 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$

4 The column space of A contains all combinations $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2$ of the columns.

5 Rank one matrices: All columns of A (and all combinations $A\mathbf{x}$) are on one line.

Square Matrices and Types

- A typical matrix A is a rectangle of m times n numbers— m rows and n columns.
- If m equals n then A is a “square matrix”.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

**Identity
matrix**

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

**Diagonal
matrix**

$$\begin{bmatrix} 2 & 1 & -3 \\ 0 & 4 & 7 \\ 0 & 0 & 5 \end{bmatrix}$$

**Triangular
matrix**

$$\begin{bmatrix} 2 & 1 & -3 \\ 1 & 4 & 7 \\ -3 & 7 & 5 \end{bmatrix}$$

**Symmetric
matrix**

Two ways to the same answer

The row picture of $A\mathbf{x}$ will come from dot products of $\mathbf{x}$ with the rows of A .

The column picture will come from linear combinations of the columns of A .

We often think of the columns of A as vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$. Each of those n vectors is in m -dimensional space. In this example the $\mathbf{a}$'s have $m = 3$ components each:

$m = 3$ rows
 $n = 4$ columns
3 by 4 matrix A

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

Row picture of $A\mathbf{x}$ Each row of A multiplies the column vector $\mathbf{x}$. Those multiplications *row times column* are dot products ! The first dot product comes from row 1 of A :

$$(\text{row } 1) \cdot \mathbf{x} = (-1, 1, 0, 0) \cdot (x_1, x_2, x_3, x_4) = \mathbf{x}_2 - \mathbf{x}_1.$$

It takes m times n small multiplications to find the $m = 3$ dot products that go into $A\mathbf{x}$.

Three rows
Three dot products

$$A\mathbf{x} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \text{row } 1 \cdot \mathbf{x} \\ \text{row } 2 \cdot \mathbf{x} \\ \text{row } 3 \cdot \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_2 - \mathbf{x}_1 \\ \mathbf{x}_3 - \mathbf{x}_2 \\ \mathbf{x}_4 - \mathbf{x}_3 \end{bmatrix} \quad (1)$$

Notice well that each row of A has the same number of components as the vector $\mathbf{x}$. Four columns of A multiply x_1 to x_4 . Otherwise multiplying $A\mathbf{x}$ would be impossible.

Column picture of $A\mathbf{x}$ The matrix A times the vector $\mathbf{x}$ is a combination of the columns of A . The n columns are multiplied by the n numbers in $\mathbf{x}$. Then add those column vectors $x_1\mathbf{a}_1, \dots, x_n\mathbf{a}_n$ to find the same vector $A\mathbf{x}$ as in equation (1):

$$A\mathbf{x} = x_1(\text{column } \mathbf{a}_1) + x_2(\text{column } \mathbf{a}_2) + x_3(\text{column } \mathbf{a}_3) + x_4(\text{column } \mathbf{a}_4) \quad (2)$$

This combination of n columns involves exactly the same multiplications as dot products of $\mathbf{x}$ with the m rows. But it is higher level! We have a vector equation instead of three dot products. You see the same result $A\mathbf{x}$ in equation (1) above and equation (3) below.

Combination of columns

$$A\mathbf{x} = x_1 \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{x}_2 - \mathbf{x}_1 \\ \mathbf{x}_3 - \mathbf{x}_2 \\ \mathbf{x}_4 - \mathbf{x}_3 \end{bmatrix} \quad (3)$$

Independence, Dependence, and Column space

Example 1

Independent columns

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix}$$

Each column gives a new direction.

Their combinations fill 3D space $\mathbf{R}^3$.

Example 2 Dependent columns

$$A_2 = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 5 \\ 6 & 0 & 6 \end{bmatrix}$$

Column 1 + column 2 = column 3
Their combinations don't fill 3D space

$$\begin{array}{l} 1 + 2 = 3 \\ 1 + 4 = 5 \\ 6 + 0 = 6 \end{array}$$

Independence, Dependence, and Column space

That plane is the column space of this matrix : Plane = all combinations of the columns.

Dependent columns in Example 2

column 1 + column 2 – column 3 is $(0, 0, 0)$.

Example 3 $A_3 = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 5 & 15 & 20 \end{bmatrix}$

Now $\mathbf{a}_2$ is 3 times $\mathbf{a}_1$. And $\mathbf{a}_3$ is 4 times $\mathbf{a}_1$.

Every pair of columns is dependent.